

24V Solenoid Valve Driver Module

Purpose: Designed a 24V Solenoid Valve Driver Module to better understand the electronic hardware used in irrigation systems and gain hands-on experience with actuator control, protection circuitry, and PCB design. The project included requirements definition, component selection, schematic capture, PCB layout, and design documentation, resulting in a complete low-side MOSFET driver solution for 24V irrigation solenoid valves.

Requirement Category	Requirement	Target Value
Input Power	Operating Input Voltage	24 VDC
Control Interface	Logic Input Compatibility	3.3V/ 5V
Output Load	Supported Load Type	24V Irrigation Solenoid Valve
Switching Method	Output Topology	Low-Side N-MOSFET
Circuit Protection	Flyback Suppression & Reverse Polarity Protection	Required

Table 1. Project Requirements

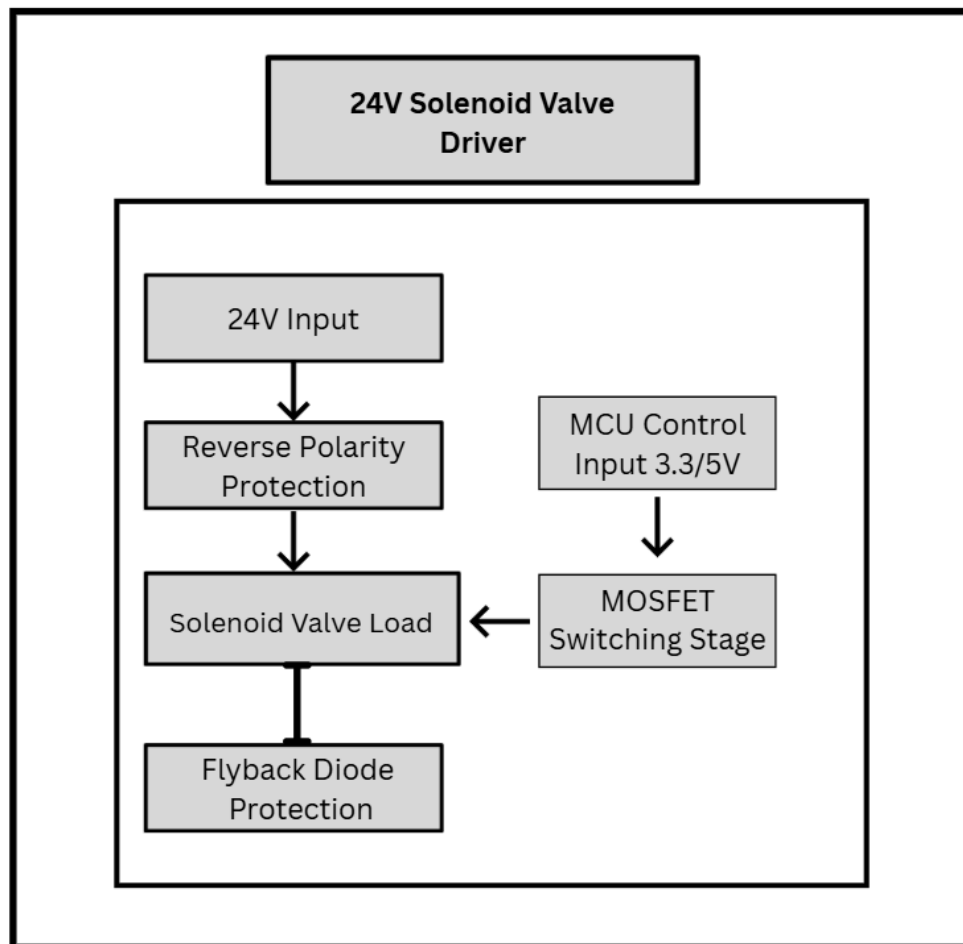


Figure 1. System Architecture Diagram

Design Highlights:

- 24V irrigation solenoid valve interface
- AO3400A logic-level N-MOSFET low-side switch
- 3.3V / 5V MCU compatible control input
- Reverse-polarity protection diode
- Flyback suppression diode
- Gate pull-down resistor for safe startup
- 2-layer PCB implementation with mounting provisions

Decision	Design Rationale
AO3400A MOSFET	Logic-level gate operation from 3.3V/5V MCU
Low-side switching topology	Simple implementation and low component count
Flyback diode	Protects MOSFET from inductive kickback
Reverse-polarity diode	Prevents damage from wiring errors
Gate pull-down resistor	Ensures MOSFET remains off during startup

Table 2. Design Decisions Rationale

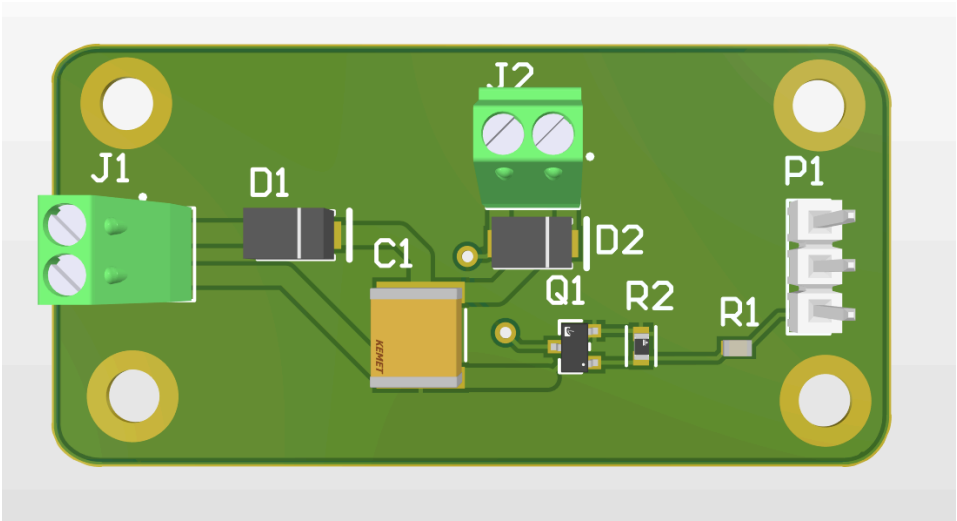


Figure 2. 3D PCB Layout Render

PCB Layout Considerations:

- Kept high-current solenoid current paths short and direct
- Positioned flyback diode physically close to solenoid connector
- Placed decoupling capacitor near protected supply node
- Used dedicated mounting holes for mechanical integration
- Routed logic-control traces separately from power paths
- Included connectorized interfaces for power, control, and load connections

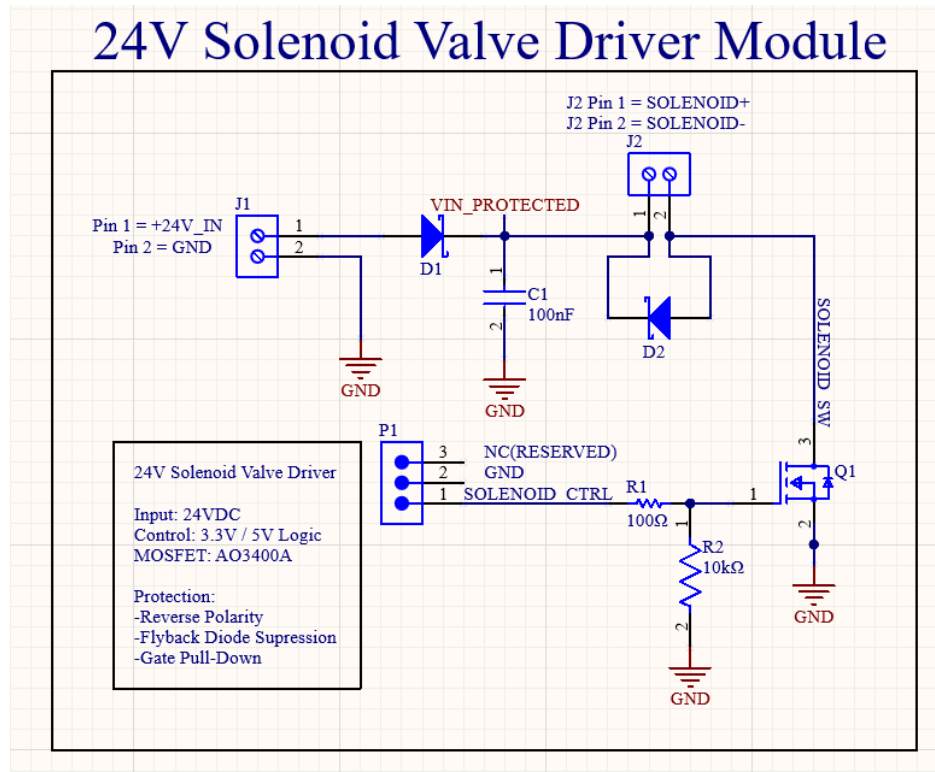


Figure 3. Low-side MOSFET-based 24V solenoid valve driver with reverse-polarity and flyback protection.

Improvement	Benefit
TVS diode protection	Increased surge robustness
Current sensing	Valve diagnostics and fault detection
PWM hold-current control	Reduced power consumption
Status LED indicators	Simplified Debugging

Table 3. Future Design Improvements

Lessons Learned: This project reinforced the importance of following a structured hardware design process, beginning with requirements definition before moving into component selection, schematic capture, and PCB layout. Through the design of a 24V solenoid valve driver, I gained a deeper understanding of how inductive loads are controlled in irrigation systems and the protection circuitry required to ensure reliable operation. The project highlighted the role of flyback suppression in protecting switching devices from inductive kickback, the importance of reverse-polarity protection for field-installed equipment, and the value of selecting components that are compatible with standard 3.3V and 5V control interfaces. Additionally, the PCB layout process emphasized the need for short high-current paths, proper component placement, mechanical mounting considerations, and clear separation between power and control circuitry. Overall, the project strengthened my understanding of actuator control hardware and provided practical experience with design decisions commonly encountered in industrial and irrigation electronics.